Combining Supervised and Unsupervised Ensembles for Knowledge Base Population

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INTRODUCTION

- In our past work, we demonstrated Stacking with auxiliary features (SWAF) for improving performance of slot filler systems[1]
- Stacking uses supervised learning and thus relies on training data
- Sometimes, we might not have access to individual systems and only to the system outputs
- We first use unsupervised ensembling to combine such systems without training data and then use stacking to combine the supervised ensemble and the unsupervised ensemble
- We obtain state-of-the-art results on two KBP tasks for 2015:

1. **Cold Start Slot Filling (CSSF)**
2. **Tri-lingual Entity Discovery and Linking (TEDL)**

OVERVIEW OF THE KBP TASKS

1. Cold Start Slot Filling (CSSF)

- Extract information (fills) about specific attributes (slots) for a set of entities(queries) from a given corpus
- Query entities can be PER/ORG/GPE

org: Microsoft
city_of_headquarters Redmond **website** microsoft.com **subsidiaries** Skype Nokia

org: Microsoft
<eng-NG-31-1007>: Microsoft is a technology company, headquartered in Redmond, Washington that develops…
city_of_headquarters Redmond **Doc ID** eng-NG-31-1007 **Start Offset** 48 **End Offset** 54

2. Tri-lingual Entity Discovery and Linking (TEDL)

- Find all entity mentions in a corpus of English, Chinese and Spanish documents and link to a FreeBase KB
- If there is not KB entry for the entity, systems are expected to cluster all the mentions for that entity using a NIL ID
- Entities can either be PER/ORG/GPE/LOC/FAC

per: Hillary Clinton
Corpus Document <doc id="“ENG_NW_001429.nw.xml”"> : Hillary Clinton Not Talking About ’92 Clinton-Gore Confederate Campaign Button **FreeBase** <ID= m.0d06m >: Hillary Diane Rodham Clinton is a US Secretary of State, U.S. Senator, and First Lady of the United States. From 2009 to 2013, she was the 67th Secretary of State, serving under President Barack Obama. She previously represented New York in the U.S. Senate. Before that, as the wife of President Bill Clinton, she was First Lady from 1993 to 2001.

ENSEMBLING ALGORITHM

- For systems common between 2014 and 2015, we use stacking to aggregate the outputs
- For systems without training data, we use constrained optimization approach described in Weng et al. to aggregate confidence scores
- Use the aggregated unsupervised ensemble as one more system to the stacker
- Unsupervised ensemble from 2014 is used as training data for this new aggregated system
- Instances classified as correct are kept and post-processed
- The final output is made to look like its produced by a single system by resolving any conflicts

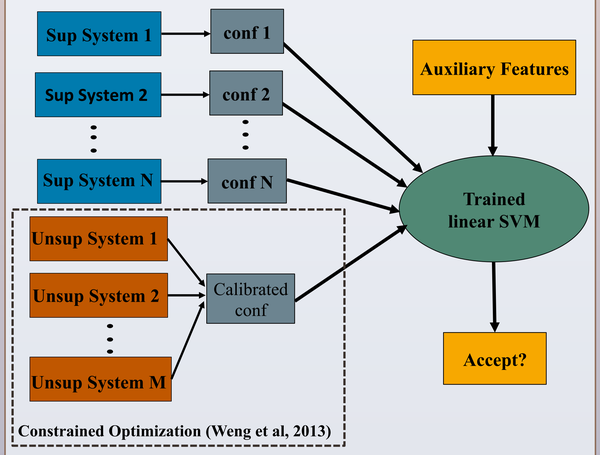

Figure 1: Illustration of our model to ensemble supervised and unsupervised systems using SWAF

	Number of supervised systems (N)			Number of unsupervised systems (M)		
	English	Chinese	Spanish	English	Chinese	Spanish
TEDL	6	4	5	4	3	2
CSSF	10	-	-	13	-	-

Table 1: Number of supervised and unsupervised systems for each of the tasks and languages

RESULTS

Approach	Precision	Recall	F1
Oracle Voting Baseline (>= 3)	0.4384	0.2720	0.3357
Top ranked CSSF system in 2015 (Angeli et al.)	0.3989	0.3058	0.3462
Stacking approach by Rajani et al.	0.4656	0.3312	0.3871
Combining supervised and unsupervised ensembles	**0.4679**	**0.4314**	**0.4489**

Table 2: Results on the 2015 CSSF task using the official NIST scorer

Approach	Precision	Recall	F1
Oracle Voting Baseline (>= 4)	0.514	0.601	0.554
Top ranked TEDL system in 2015 (Sil et al.)	0.693	0.547	0.611
Stacking approach by Rajani et al.	**0.813**	0.515	0.630
Combining supervised and unsupervised ensembles	0.686	**0.624**	**0.653**

Table 3: Results on the 2015 TEDL task using the official NIST scorer and CEAF metric[6]

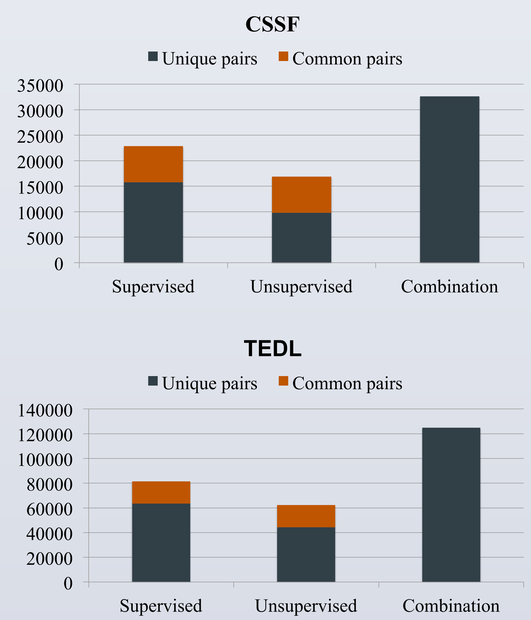

Figure 2: Proportion of unique and common input pairs in the ensemble for the CSSF and TEDL tasks

CONCLUSIONS

- Stacking-based approach to ensembling both supervised and unsupervised systems is very promising on CSSF and TEDL tasks
- Our model outperforms top ranked systems from the 2015 competition as well as several other ensembling methods on both tasks
- We obtained new state-of-the-art results for CSSF and TEDL
- Adding the unsupervised ensemble along with the shared systems leads to a substantial increase in recall

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